

Mark Walker Memorial Ground Station Competition 2027

Rev 1.0

Introduction

The Mark Walker Portable Ground Station Design Award is presented to the team that designs and builds the best custom, fully functioning portable ground station. Participation is not mandatory but is encouraged. Craftsmanship, artistry, and creativity are encouraged and rewarded.

Mark Walker was a long-time CanSat team member who specialized in ground station development and command and telemetry operations for satellite systems. He developed the requirements for the ground station, telemetry, and commanding used in the competition.

Summary

Teams shall design a portable ground station that can be carried by one person and weighs no more than 15 pounds. The ground station shall be a single hand-carried device requiring no assembly to be functional, other than attaching an antenna. It may be enclosed to protect it from the environment — built into a commercial case (e.g., a Pelican-style case), a custom enclosure, or a fully custom chassis.

The ground station shall be able to be placed on a table, opened, connected to an antenna, and be ready to operate when powered. The ground station software shall start automatically on power-up with no operator intervention.

The ground station must support the full 2027 mission: simultaneous reception, display, and logging of Container and PocketQube telemetry, and transmission of all required commands.

Design Criteria

Item	Description
1	Must not use a laptop or repackage a laptop.
2	Keyboard and mouse disallowed.
3	Must be a single package, excluding the antenna.
4	May use a commercial case or a custom-built enclosure.
5	Must not require assembly except for attaching an antenna.
6	Must weigh under 15 pounds.
7	Must use only one switch and/or button to power on and off.
8	Ground station software shall start automatically when powered.
9	Must contain its own power source (e.g., battery) sufficient for the full flight-day operational window.
10	Must perform all required ground station requirements (GS1–GS4 and all command/telemetry requirements of the mission guide).
11	Cannot be remotely controlled or remotely operated.
12	Must allow a USB memory device to be connected for capturing telemetry.
13	Must include all needed user input devices.
14	Must include a display or displays showing Container and PocketQube telemetry (GS1).
15	Must include the radio used to communicate with the CanSat.

Judging Process

Judging takes place in three stages:

1. **Stage 1 — Blind Judging (teams absent).** Judges inspect the powered-off ground stations without team members present. Physical compliance, construction quality, craftsmanship, and theme are scored. This stage rewards work that speaks for itself.
2. **Stage 2 — Guided Demonstration (teams present).** Each team powers on and demonstrates its ground station using simulated or test telemetry. Judges score power-up behavior, display completeness, commanding, and usability.
3. **Stage 3 — Flight Day.** The ground station must perform during the actual flight. Judges verify telemetry reception, live plotting, in-flight commanding, and data capture. This is captured in the Flight Day scoring. Mark Walker stations must be used during Flight Day operations or the team won't be eligible for the award.

Scoring Key

- **Yes/No items:** Yes= present and fully functional · No = absent / not fully functional.
- **0-5 items:** Judge's discretion on quality. 0 = no effort · 1-2 = minimal · 3 = solid · 4 = excellent · 5 = exceptional / memorable.

Team Number:	Judge:	Team Initial:
Item	Result	
Blind Judging		
Single package	Yes	No
Not a laptop/repackaged laptop	Yes	No
Weight < 15 lbs / 6.8 kg	Yes	No
Single power switch	Yes	No
Self-contained power source	Yes	No
USB port for telemetry is present and accessible	Yes	No
All user input devices present	Yes	No
All displays are connected and present	Yes	No
Construction quality — panel fit, enclosure fit, solid construction, nothing loose or easily deformed, no holes	0 1 2 3 4 5	
Craftsmanship — custom fabrication, finish, attention to detail	0 1 2 3 4 5	
Theme & coolness — is there a theme, and is it carried through the whole build?	0 1 2 3 4 5	
Controls - Are they integrated into the design/aesthetic? Was there thought put into the implementation?	0 1 2 3 4 5	
Display(s) - Is/are they noteworthy? Has thought been put into visibility?	0 1 2 3 4 5	
Total Score	/ 41	

Team Number:	Judge:	Team Initial:
Item	Result	
Demo Judging		
Single package	Yes	No
Not a laptop/repackaged laptop	Yes	No
Weight < 15 lbs / 6.8 kg	Yes	No
Single power switch	Yes	No
Self-contained power source	Yes	No
USB port for telemetry is present and accessible	Yes	No
All user input devices present	Yes	No
All displays are connected and present	Yes	No
Powers on with a single switch/button	Yes	No
Software starts automatically and reaches the operational dashboard with no operator intervention	Yes	No
Ready to operate (receiving-capable) within a reasonable startup time	Yes	No
User interface — Display(s) easy to understand; all required plots shown and easily identified; control interface well organized	0 1 2 3 4 5	
Ease of use — physical input devices labeled/identifiable; an unfamiliar operator could run it; USB save is straightforward	0 1 2 3 4 5	
Display quality — Display(s) theme is unique, legible, and consistent with the build's style	0 1 2 3 4 5	
Total Score	/ 21	